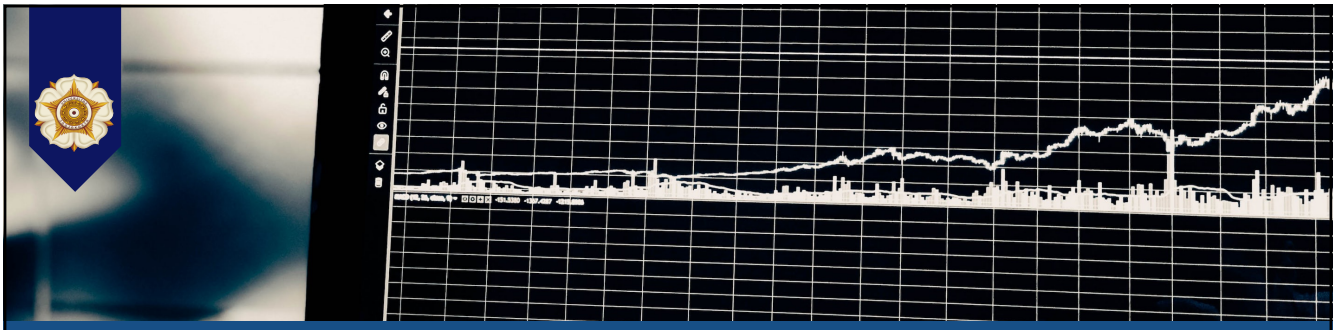


Mathematical Estimation and Discrete Probability Distribution

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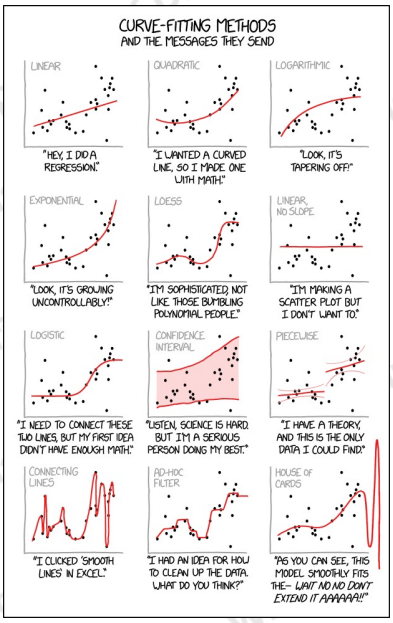
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1

Overview of today's lecture

- Mean of random variable
- Variance, covariance, correlation coefficient
- Uniform and binomial distribution
- Normal distribution



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2

Part 1

MEAN OF RANDOM VARIABLE

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Random variable (review)

- **Probability theory:** the discipline concerned with the study of uncertain (or random) phenomena. Probability is the mathematical language adopted for quantifying uncertainty.
- Such phenomena, although not predictable in a deterministic fashion, may present some regularities and consequently be described mathematically by idealized probabilistic models.
- The theory of probability makes possible to infer from these models the patterns of future behavior.
- Random variable is useful to express random quantities determined by outcome of experiment

Table 4-1
Probabilities of Girls

x (girls)	$P(x)$
0	0.000
1	0.001
2	0.006
3	0.022
4	0.061
5	0.122
6	0.183
7	0.209
8	0.183
9	0.122
10	0.061
11	0.022
12	0.006
13	0.001
14	0.000

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Experiment with coins

- Two coins are tossed 16 times and X is the number of heads that occur per toss, then the values of X can be 0, 1, and 2.
- Suppose the experiment yields no heads, one head, and two heads a total of 4, 7, and 5 times, respectively. The average number of heads per toss of the two coins is:

$$(0)\left(\frac{4}{16}\right)+(1)\left(\frac{7}{16}\right)+(2)\left(\frac{5}{16}\right)=1.06$$

- 1.06 is the mean of random variable X, or the mean of the probability distribution of X, or mathematical expectation / expected value of random variable X

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Expected value

Let *X* be a random variable with probability distribution *f*(*x*). The **mean** or **expected value** of X is

$$\mu = E(x) = \sum_x x f(x) \qquad \text{If } X \text{ is discrete}$$

$$\mu = E(x) = \int_{-\infty}^{\infty} x f(x) dx \qquad \text{If } X \text{ is continuous}$$

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Comparison of mean formulas

- General formula of mean:

$$\bar{x} = \sum_{i=1}^{n} \frac{x_i}{n} = \sum_{x} x \left(\frac{1}{n}\right)$$

- Discrete Expected Value

$$\mu = E(X) = \sum_{x} x\, f(x)$$

- Continuous Expected Value

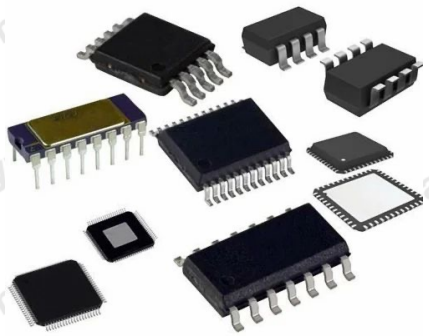
$$\mu = E(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx$$

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7

Example

A lot containing 7 components is sampled by a quality inspector. The lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector.



Find the expected value of the number of good components in this sample.

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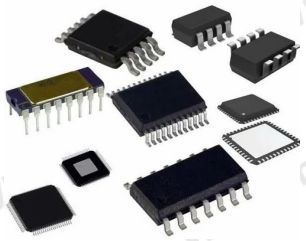
Answer

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of **the number of good components** in this sample.

Solution

- Let *X* represent the number of good components in the sample. The probability distribution of *X* is

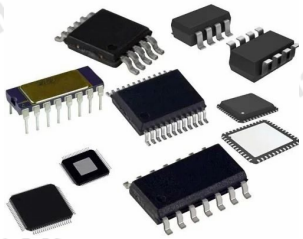
$$f(x) = \frac{\binom{4}{x}\binom{3}{3-x}}{\binom{7}{3}}, \qquad x = 0, 1, 2, 3.$$



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9

Answer



A few simple calculations yield
f(0) = 1/35, *f*(1) = 12/35, *f*(2) = 18/35, *f*(3) = 4/35.

Therefore:

$$\mu = E(X) = (0)\frac{1}{35} + (1)\frac{12}{35} + (2)\frac{18}{35} + (3)\frac{4}{35}$$
$$= 1.7$$

Thus, if a sample of size 3 is selected at random over and over again from a lot of 4 good components and 3 defective components, it would contain, on average, **1.7 good components.**

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10

Example

Let *X* be the random variable that denotes the life in days of a smart light bulb. The probability density function is

$$f(x) = \begin{cases} \frac{20{,}000}{x^3}, & x > 100 \\ 0, & \text{elsewhere} \end{cases}$$



Find the expected life of the smart light bulb.

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Answer

We use this integral formula: $\int ax^n \, dx = a\frac{x^{n+1}}{n+1}, n \neq -1$

$$\mu = E(X) = \int_{100}^{\infty} x\frac{20,000}{x^3}dx = \int_{100}^{\infty} \frac{20,000}{x^2}dx$$
$$= 200$$



Therefore, we can expect the smart light bulb to last, *on average,* 200 days (approx. 6.5 months).

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12

Mean of g(X)

Let X be a random variable with probability distribution $f(x)$. The mean or expected value of the random variable g(X) is

$$\mu_{g(X)} = E[g(X)] = \sum_x g(x)f(x) \qquad \text{If } X \text{ is discrete}$$

$$\mu_{g(X)} = E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx \qquad \text{If } X \text{ is continuous}$$

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13

Example

Suppose that the number of cars X that pass through a car wash between 4:00 P.M. and 5:00 P.M. on any sunny Friday has the following probability distribution:

x	4	5	6	7	8	9
P(X=x)	1/12	1/12	1/4	1/4	1/6	1/6



Let $g(X) = 2X - 1$ represent the amount of money in **dollars**, paid to the attendant by the manager. Find the attendant's expected earnings for this particular time period.

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14

Answer



$$
\begin{aligned}
E[g(X)] &= E(2X-1) \\
&= \sum_{x=4}^{9}(2x-1)f(x) \\
&= (7)\frac{1}{12} + (9)\frac{1}{12} + (11)\frac{1}{4} + (13)\frac{1}{4} + (15)\frac{1}{6} + (17)\frac{1}{6} \\
&= 12.67
\end{aligned}
$$

The attendant's expected earnings for this particular time period is 12.67 USD.

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15

Part 2

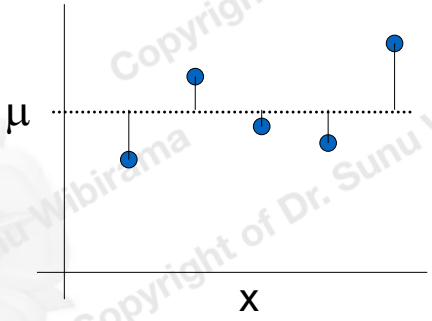
VARIANCE, COVARIANCE, CORRELATION COEFFICIENT

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16

Variance

- Mean denotes where the probability distribution is centered
- Mean does not give adequate description of "shape of distribution"
- We need to characterize the variability in the distribution



Reminder: variance of population

$$\sigma^2 = \sum_{i=1}^{n} \frac{(x_i - \mu)^2}{N}$$

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17

Variance

Let X be a random variable with probability distribution $f(x)$ and mean μ.
The **variance of** X is

$$\sigma^2 = E[(X-\mu)^2] = \sum_x (x-\mu)^2 f(x) \quad \text{If } X \text{ is discrete}$$

$$\sigma^2 = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)dx \quad \text{If } X \text{ is continuous}$$

The **standard deviation of x** is the positive square root of the variance of x:

$$\sigma = \sqrt{\sigma^2}$$

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Example

Refer to the two-coin tossing experiment and the probability distribution for the random variable *x*. Find the variance and standard deviation of random variable *x*.



S={(T,T), (T,H), (H, T), (H, H)}

Solution

If we were to repeat the two-coin tossing experiment a large number of times – say 400,000 times, we would expect to observe:

x = 0 heads approximately 100,000 times,
x = 1 head approximately 200,000 times,
x = 2 heads approximately 100,000 times.

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19

Example

Calculating the mean of these 400,000 values of *x*, we obtain

$$\mu \approx \frac{\sum_x x}{N} = \frac{100,000(0) + 200,000(1) + 100,000(2)}{400,000}$$
$$= \frac{1}{4}(0) + \frac{1}{2}(1) + \frac{1}{4}(2)$$
$$= \sum_x f(x)x$$

Thus, the mean of *x* is 1
The average numbers of head per toss is 1

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20

Example (continued)

Variance of random variable x

$$\sigma^2 = E[(x-\mu)^2] = \sum_{x=0}^{2}(x-\mu)^2 p(x)$$
$$= (0-1)^2\left(\frac{1}{4}\right) + (1-1)^2\left(\frac{1}{2}\right) + (2-1)^2\left(\frac{1}{4}\right)$$
$$= \frac{1}{2}$$

Standard deviation:

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{2}} \approx 0.707$$

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21

Covariance

- Covariance: measurement of the nature association between two dependent random variables X and Y
- If covariance sign is positive (+), then $X \sim Y$
- If covariance sign is negative (-), then $X \sim \frac{1}{Y}$
- If X and Y is statistically independent, then covariance equals to 0
- Covariance of X and Y :

$$\sigma_{XY} = E[(X-\mu_X)(Y-\mu_Y)] \quad \text{or} \quad \sigma_{XY} = E(XY) - \mu_X\mu_Y$$

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Example

Two refills for a ballpoint pen are selected at random from among 3 blues, 2 reds, and 3 greens. If X is the number of blue refills and Y is the number of red refills selected, find the covariance of X and Y.

Solution:

Possible pairs of numbers of blue & red are: (0,0), (0,1), (1,0), (1,1), (0,2), (2,0)

(0,0) means : no blue, no red (but there is possibility that two green pens are selected)

(0,1) means: no blue, 1 red (but there is possibility that one green pen is selected)

(1,1) means : 1 red, 1 blue

Now, $f(0,1)$, for example, represents the probability that a red and a green pens are selected. The total number of equally likely ways of selecting any 2 pens from the 8 is $\binom{8}{2} = 28$. The number of ways of selecting 1 red from 2 red pens and 1 green from 3 green pens is $\binom{2}{1}\binom{3}{1} = 6$. Hence, $f(0,1) = 6/28 = 3/14$. Similar calculations yield the probabilities for the other cases, which are presented in Table 3.1. Note that the probabilities sum to 1.



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Example

- Given the following probability density distribution, find the covariance of *x* and *y*:

		x			
f(x,y)		0	1	2	Row totals
y	0	3/28	9/28	3/28	15/28
	1	6/28	6/28		12/28
	2	1/28			1/28
Col. totals		10/28	15/28	3/28	1

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Example (continued)

f(x,y)	x = 0	1	2	h(y)
y = 0	3/28	9/28	3/28	15/28
1	6/28	6/28		12/28
2	1/28			1/28
g(x)	10/28	15/28	3/28	1

Step 1:
Find the mean of X and Y

$$\mu_x = E(X) = \sum_{x=0}^{2}\sum_{y=0}^{2} x\, f(x,y) = \sum_{x=0}^{2} x\, g(x)$$

$$= 0(10/28) + 1(15/28) + 2(3/28) = 3/4$$

$$\mu_y = E(Y) = \sum_{x=0}^{2}\sum_{y=0}^{2} y\, f(x,y) = \sum_{y=0}^{2} y\, h(y)$$

$$= 0(15/28) + 1(12/28) + 2(1/28) = 14/28 = 1/2$$

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25

Example (continued)

f(x,y)	x = 0	1	2	h(y)
y = 0	3/28	9/28	3/28	15/28
1	6/28	6/28		12/28
2	1/28			1/28
g(x)	10/28	15/28	3/28	1

Step 2: find the covariance of X and Y

$$E(XY) = \sum_{x=0}^{2}\sum_{y=0}^{2} x\, y\, f(x,y)$$

$$= 0(0)(3/28) + 0(1)(6/28) + 0(2)(1/28) + (1)(0)(9/28) + 1(1)(6/28) + 2(0)(3/28)$$

$$= 3/14$$

$$\sigma_{XY} = E(XY) - \mu_x\mu_y = 3/14 - (3/4)(1/2) = -9/56$$

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Correlation coefficient

- Note that value of co-variance does not indicate strength of the relationship since it depends on the scale. The correlation coefficient is a scale-free version.
- Let X and Y be random variables with standard deviations σ_X and σ_Y, and covariance σ_{XY} . The correlation coefficient of X and Y is:

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} \qquad -1 \le \rho_{XY} \le 1$$

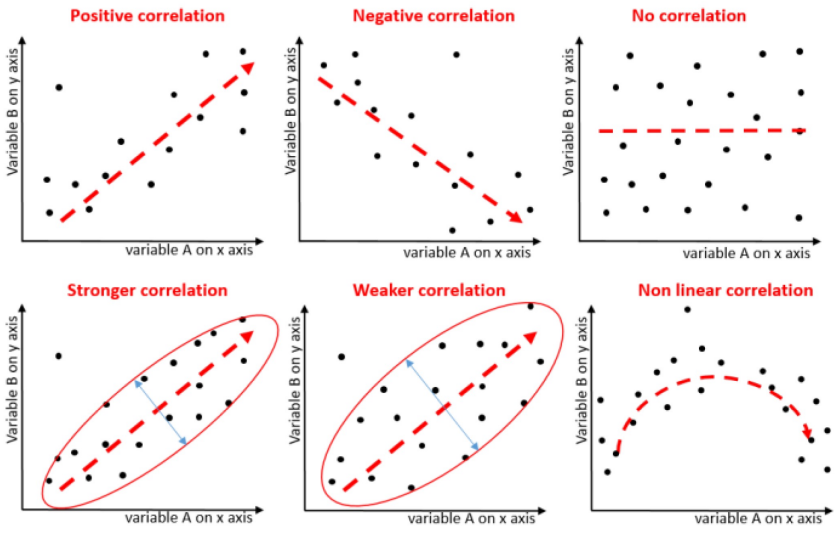
The correlation coefficient is scale-free, and:

$$if\ \ \sigma_{XY} = 0, \text{then } \rho_{XY} = 0$$

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Correlation



Kumar, S.; Chong, I. Correlation Analysis to Identify the Effective Data in Machine Learning: Prediction of Depressive Disorder and Emotion States. *Int. J. Environ. Res. Public Health* **2018**, *15*, 2907. https://doi.org/10.3390/ijerph15122907

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Correlation vs Covariance

Covariance	Correlation
• Indicates direction of linear relationship	• Indicates direction and strengt of linear relationship
• Positive covariance indicates an increase in one variable indicates an increase in the other	• Positive correlation coeff close to 1 indicates a strong positive corr and value close to -1 indicates strong nevative corr
• Covariance can be between -infinity to infinity	• Correlation coefficient can be between -1 and 1

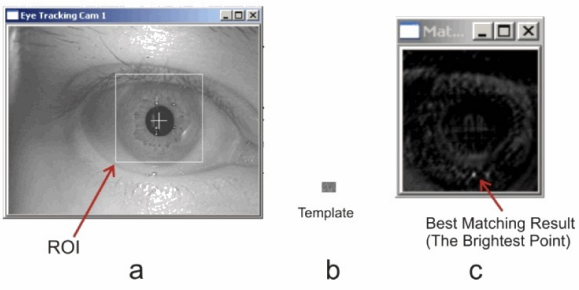
Kumar, S.; Chong, I. Correlation Analysis to Identify the Effective Data in Machine Learning: Prediction of Depressive Disorder and Emotion States. *Int. J. Environ. Res. Public Health* **2018**, *15*, 2907. https://doi.org/10.3390/ijerph15122907

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29

Real implementation

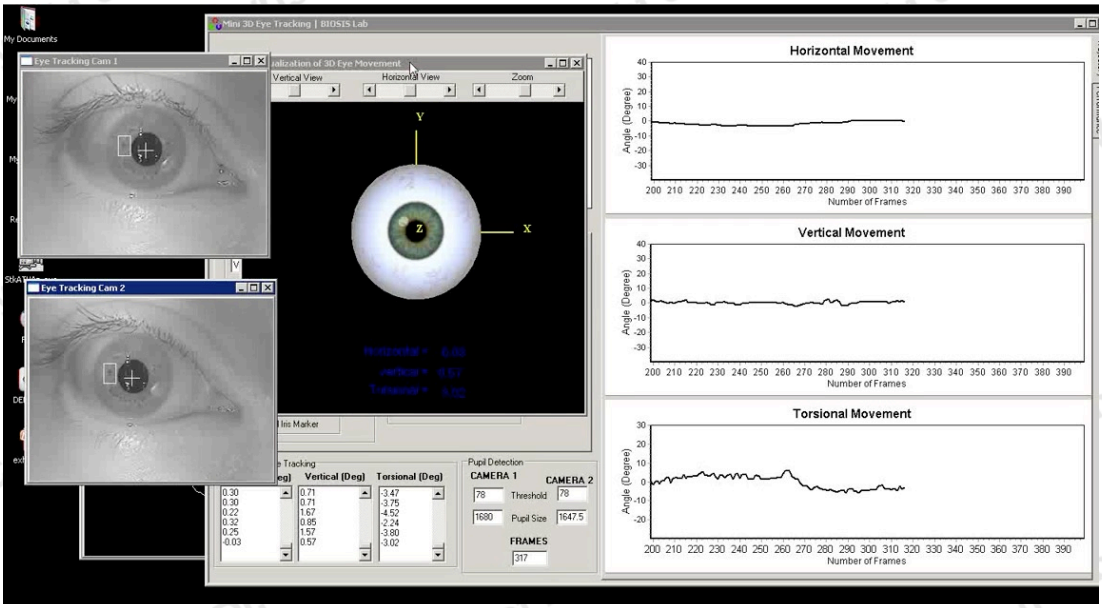
- Correlation coefficient can be used in “template matching algorithm” for pattern recognition or object tracking
- Find the position of small patch of an image, in the sequential frames. Best matching will be the maximum value of correlation coefficient



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30

Real implementation



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31

Part 3

UNIFORM AND BINOMIAL DISTRIBUTION

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32

Discrete uniform distribution

If the random variable *X* assume the values $x_1, x_2, \ldots, x_k$, with **equal probabilities**, then the discrete uniform distribution is given by

$$f(x;k) = \frac{1}{k}, \quad x = x_1, x_2, \ldots, x_k$$

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Example

When a light bulb is selected at random from a box that contains a 40-watt bulb, a 60-watt bulb, a 75-watt bulb, and a 100-watt bulb, each element of the sample space *S* = {40, 60, 75, 100} occurs with probability 1/4. Therefore, we have a uniform distribution, with

$$f(x;4) = \frac{1}{4}, \quad x = 40, 60, 75, 100$$



When a fair die is tossed, each element of the sample space *S* = {1, 2, 3, 4, 5, 6} occurs with probability 1/6. Therefore, we have a uniform distribution, with

$$f(x;6) = \frac{1}{6}, \quad x = 1, 2, 3, 4, 5, 6$$



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Mean and variance of discrete uniform distribution

The mean and variance of the discrete uniform distribution $f(x;k)$ are

$$\mu = \frac{\sum_{i=1}^{k} x_i}{k}$$

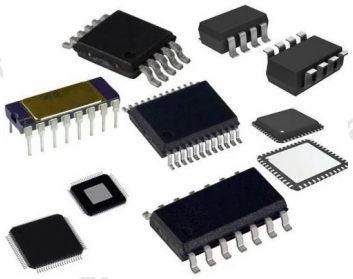
$$\sigma^2 = \frac{\sum_{i=1}^{k} (x_i - \mu)^2}{k}$$

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35

Binomial distribution

- An experiment often consists of repeated trials, each with two possible outcomes that may be labeled **success** or **failure**.
- The most obvious application deals with the testing of items as they come off an assembly line, where each test or trial may indicate a **defective** or a **non-defective** item (or coin, with head or tail)
- We may choose to define either outcome as a **success**. The process is referred to as a **Bernoulli process**. Each trial is called a **Bernoulli trial**.

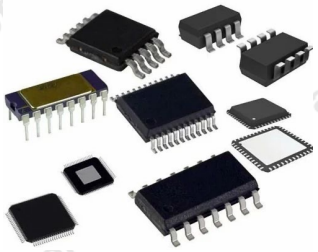


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Important: characteristics of a binomial random variable

- The experiment consists of *n* identical trials
- There are only 2 possible outcomes on each trial. We will denote one outcome by *S* (for **Success**) and the other by *F* (for **Failure**).
- The probability of *S* **remains the same from trial to trial**. This probability will be denoted by *p*, and the probability of *F* will be denoted by *q* (*q* = 1-*p*).
- The trials are independent.
- The binomial random variable *X* is the number of *S* in *n* trials.



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Probability distribution of binomial random variable

A Bernoulli trial can result in a success with probability *p* and a failure with probability *q* = 1 − *p.*

Then the probability distribution of the binomial random variable *X*, the number of successes in *n* independent trials, is

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, ..., n.$$

where
p = probability of a success on a single trial, *q* = *1−p*
n = number of trials, *x* = number of successes in *n* trials

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Example

Consider the set of Bernoulli trials where three items are selected at random from a manufacturing process, inspected, and classified as defective or nondefective. A defective item is designated a success. The number of successes is a random variable X assuming integral values from 0 through 3. The eight possible outcomes and the corresponding values of X are

Outcome	NNN	NDN	NND	DNN	NDD	DND	DDN	DDD
x	0	1	1	1	2	2	2	3

Since the items are selected independently and we assume that the process produces 25% defectives, we have

$$P(NDN) = P(N)P(D)P(N) = \left(\frac{3}{4}\right)\left(\frac{1}{4}\right)\left(\frac{3}{4}\right) = \frac{9}{64}.$$

Similar calculations yield the probabilities for the other possible outcomes. The probability distribution of X is therefore

x	0	1	2	3
$f(x)$	$\frac{27}{64}$	$\frac{27}{64}$	$\frac{9}{64}$	$\frac{1}{64}$

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39

Example

A Bernoulli trial can result in a success with probability p and a failure with probability $q = 1 - p$. Then the probability distribution of the binomial random variable X, the number of successes in n independent trials, is

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \ldots, n.$$

Note that when $n = 3$ and $p = 1/4$, the probability distribution of X, the number of defectives, may be written as

$$b\left(x; 3, \frac{1}{4}\right) = \binom{3}{x}\left(\frac{1}{4}\right)^x\left(\frac{3}{4}\right)^{3-x}, \quad x = 0, 1, 2, 3,$$

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40

Part 4

NORMAL DISTRIBUTION

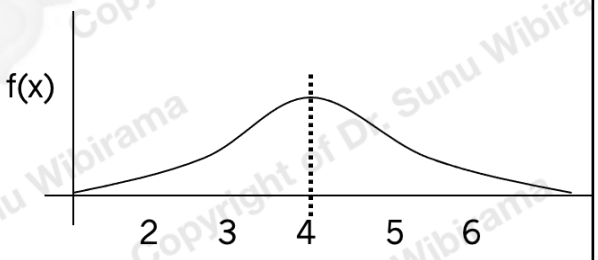
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Normal distribution

- The most important continuous probability distribution
- The graph called the “normal curve” (bell-shaped).
- Total area under the curve = 1
- Derived by DeMoivre and Gauss. The distribution is also called the **“Gaussian” distribution.** It describes many phenomena in nature, industry and research.

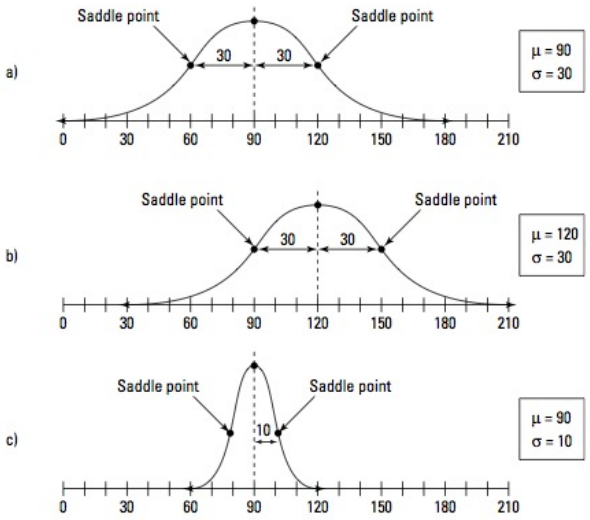
$$f(x) = n(x;\mu,\sigma) = \frac{e^{-0.5\left[\frac{(x-\mu)}{\sigma}\right]^2}}{\sigma\sqrt{2\pi}}, \quad -\infty < x < \infty$$



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42

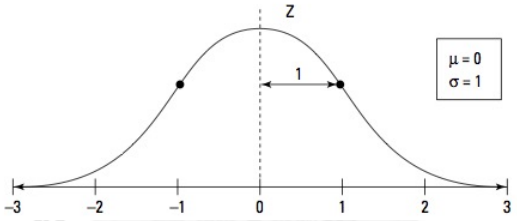
Various normal distribution



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43

Standard normal distribution

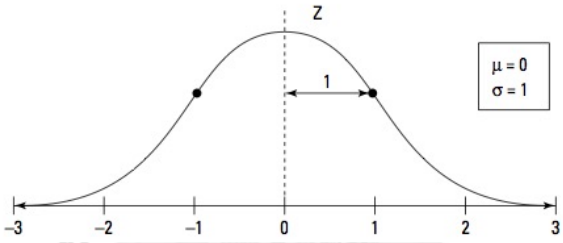


- General density function: $$f(x) = n(x;\mu,\sigma) = \frac{e^{-0.5\left[\frac{(x-\mu)}{\sigma}\right]^2}}{\sigma\sqrt{2\pi}}$$
- For **standardized** normal distribution: $$f(x) = n(x;\mu,\sigma) = \frac{e^{-\left[\frac{x}{2}\right]^2}}{\sqrt{2\pi}}$$

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44

Standard normal distribution



- It's also called **z-distribution**
- It has a mean of 0 and standard deviation of 1
- Transforming normal random variable *x* to standard normal random variable *z*

$$z = \frac{x - \mu}{\sigma}$$

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45

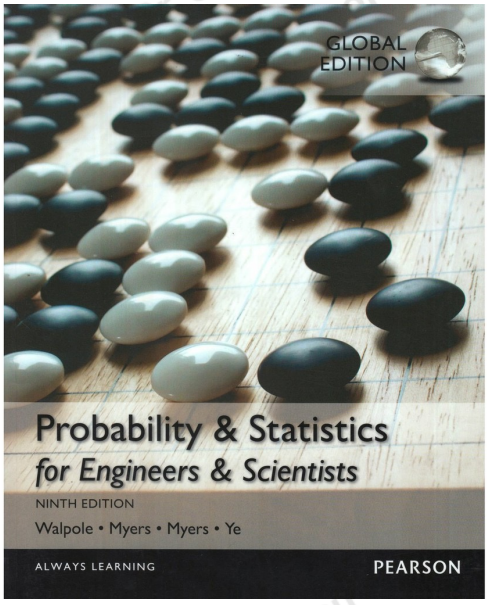
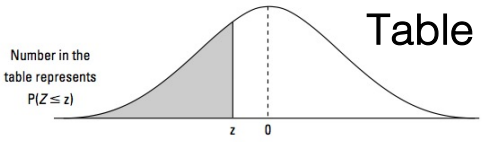


Table A.3



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−3.6	.0002	.0002	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
−3.5	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0003	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233

Find area under normal curve
Example: P (Z < -2.13) = 0.0166

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46

Solving a case using Z-distribution

- Suppose we have a problem finding probability of a case in *X* domain → *p(X)*
- We can use Z-distribution to solve the problem
- We should transform X → Z ("Finding X from Z")

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Important: Finding *x* from *z*

1. **Draw a picture of the distribution.**
2. **Translate the problem into one of the following:**

$$P(X < a), \quad P(X > b), \quad \text{or} \quad P(a < X < b).$$

Shade in the area on your picture.

3. **Transform a (and/or b) into a z-value, using the Z-formula:**

$$Z = \frac{X - \mu}{\sigma}.$$

4. **Look up the transformed z-value on the Z-table (in the Walpole's book, see the Table A.3) and find its probability.**

5a. **If you have a less-than problem, you are done.**

5b. **If you have a greater-than problem, take one minus the result from Step 4.**

5c. **If you have a between-values problem, do Steps 1–4 for b (the larger of the two values) and then for a (the smaller of the two values), and subtract the results.**

Table A.3 Normal Probability Table 755



Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

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Example

Suppose, for example, that you enter a fishing contest. The contest takes place in a pond where the fish lengths have a normal distribution with mean $\mu = 16$ inches and standard deviation $\sigma = 4$ inches.

Problem 1: Whats the chance of catching a small fish say, less than 8 inches?

Problem 2: Suppose a prize is offered for any fish over 24 inches. Whats the chance of catching a fish at least that size?

Problem 3: Whats the chance of catching a fish between 16 and 24 inches?

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49

Answer

To solve these problems, first draw a picture of the distribution. Figure 5-3 shows a picture of X's distribution for fish lengths. You can see where each of the fish lengths mentioned in each of the three fish problems falls.

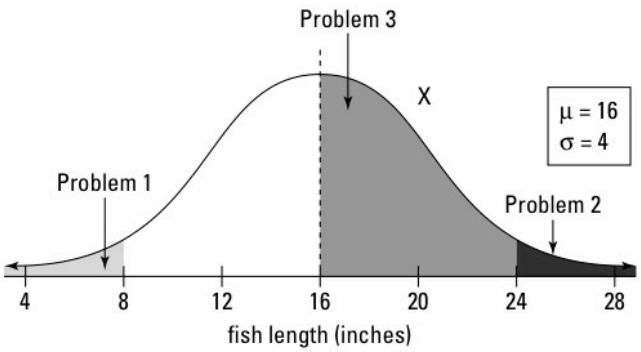


Figure 5-3: The distribution of fish lengths in a pond.

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50

Answer

Next, translate each problem into probability notation. Problem 1 means find $P(X < 8)$. For Problem 2, you want $P(X > 24)$. And Problem 3 is asking for $P(16 < X < 24)$.

Step 3 says change the x-values to z-values using the Z-formula, $Z = \frac{X - \mu}{\sigma}$. For Problem 1 of the fish example, you have $P(X < 8) = P\left(Z < \frac{8-16}{4}\right) = P(Z < -2)$. Similarly for Problem 2, $P(X > 24)$ becomes $P(Z > 2)$. Problem 3 translates from $P(16 < X < 24)$ to $P(0 < Z < 2)$. Figure 5-4 shows a comparison of the X-distribution and Z-distribution for the values $x = 8$, 16, and 24, which transform into $z = -2$, 0, and +2, respectively.

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51

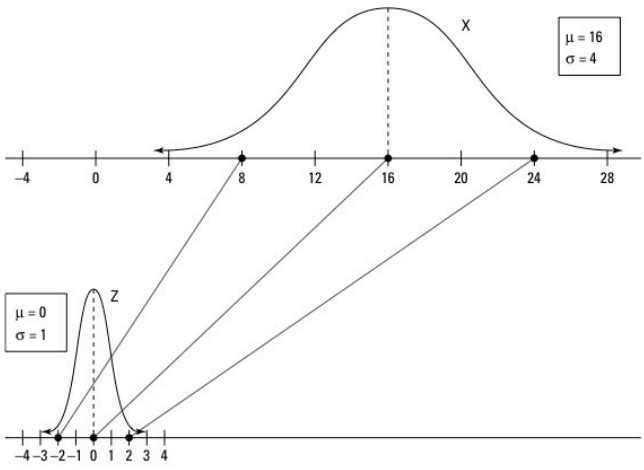
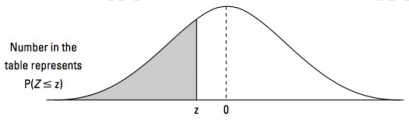


Figure 5-4: Transforming numbers on the normal distribution to numbers on the Z-distribution.



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−3.6	.0002	.0002	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
−3.5	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0003	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233

Now that you have changed x-values to z-values, you move to Step 4 and find probabilities for those z-values using the Z-table (Table A-3 in the appendix). In Problem 1 of the fish example, you want $P(Z < -2)$; go to the Z-table and look at the row for −2.0 and the column for 0.00, intersect them, and you find 0.0228 — according to Step 5a you're done. So, the chance of a fish being less than 8 inches is equal to 0.0228.

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52

Answer

For Problem 2, find P(Z > 2.00). Because it's a "greater-than" problem, this calls for Step 5b. To be able to use the Z-table you need to rewrite this in terms of a "less-than" statement. Because the entire probability for the Z-distribution equals 1, we know P(Z > 2.00) = 1 – P(Z < 2.00) = 1 – 0.9772 = 0.0228.

So, the chance that a fish is greater than 24 inches is 0.0228. (Note the answers to Problems 1 and 2 are the same because the Z-distribution is symmetric; see Figure 5-3.)

In Problem 3, you find P(0 < Z < 2.00); this requires Step 5c. First find P(Z < 2.00), which is 0.9772 from the Z-table, and then subtract off the part you don't want, which is P(Z < 0) = 0.500 from the Z-table. This gives you 0.9772 – 0.500 = 0.4772. So the chance of a fish being between 16 and 24 inches is 0.4772.

Table A.3 (continued) Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

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53

Take home messages

- In this lecture, learn about mathematical expectation that consists of mean and variance of random variable.
- We also learn about covariance and correlation coefficient that explain relationship between two random variables.
- There are various distributions of discrete random variables. We learn three of them: uniform, binomial, and normal distribution.
- Normal distribution is the most important discrete probability distribution that resembles so many natural phenomena in industry and scientific research.

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54

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55